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RESEARCH ARTICLE

Association of placental pathology and gross morphology with autism spectrum disorders

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Abstract

This study evaluated the association between placental pathology and gross morphology and the risk of autism spectrum disorders (ASD). We conducted a matched case–control study of children with confirmed ASD who were born between 2000 and 2017 at one of three university-affiliated hospitals in Montreal, Quebec. Cases, who were identified through the Montreal Children's Hospital Autism Spectrum Disorders Program, were matched to babies (1:5) born at the same hospital and on the same day. Multi-fetal births were excluded. Maternal demographics, pregnancy characteristics and placental pathologies were collected from hospital charts by abstractors blind to autism diagnoses. This current study consisted of data from a single-site that had pathology reports available. Pearson chi-square and Wilcoxon rank-sum tests were used to estimate p-values. Our study consisted of 107 ASD cases and 526 matched controls. Mothers of cases and controls were similar in terms of parity, gravidity, smoking status, BMI, rates of clinical chorioamnionitis, chronic hypertension, and gestational diabetes. Age at delivery of <25 years was more common among mothers of controls. Compared with controls, cases were more likely born male, <32 weeks of gestation, and weighing <1500 g. Cases and controls had similar rates of placental inflammation, vasculitis, and other placental pathologies. There were no differences in placental weight, placental thickness, umbilical cord length, and umbilical cord insertion between the two groups. In conclusion, placental pathology and gross morphology do not appear to be associated with ASD, suggesting that any perinatal determinants of autism are not likely to be mediated through placental pathology.

Lay Summary

Data from a matched case–control study consisting of neonates born between 2000 and 2017 at one of three McGill-affiliated hospitals were used to examine the relationship between placental pathology and morphology and the development of autism. No differences in placental pathology and gross morphology were found between those with and without autism, which suggests that placental abnormalities are unlikely to either cause or mediate the development of autism.

KEYWORDS
autism spectrum disorder, intrauterine inflammation, placenta, umbilical cord, vasculitis

INTRODUCTION

The prevalence of autism spectrum disorders (ASD) has risen in recent decades, with estimates reaching 1 in 160 children worldwide (Elsabbagh et al., 2012), and about 1 in 59 children in the United States ("National Center on Birth Defects and Developmental Disabilities, Centers for Disease Control and Prevention, 2021; Accessed on January 27, 2021"). Despite this increase in ASD, its etiology is unknown but is believed to be multifactorial in nature, with growing evidence suggesting that non-heritable pre- and perinatal influences interplay with genetic factors to increase the risk of offspring ASD (Gardener et al., 2009, 2011; Ronald & Hoekstra, 2011).

Emergent findings increasingly suggest that intrauterine inflammation and abnormal placental function may play a role in the development of ASD and other neurodevelopmental disorders, such as cerebral palsy, schizophrenia, and psychomotor impairment (Goldstein et al., 2020; Suvisaari et al., 2013; Xiao et al., 2018; Xing et al., 2019). For instance, it has been theorized that in utero immune activation could contribute to cytokine-mediated fetal inflammatory response (Tomlinson et al., 2019). Increases in pro-inflammatory cytokines could, in turn, disrupt fetal brain development. Chronic intrauterine inflammatory processes may also affect placental angiogenesis, and contribute to hypoxic–ischemic insults to the fetal brain (Armstrong-Wells et al., 2015; Kuypers et al., 2012; van Vliet et al., 2012). Alterations of white matter during gestation may ultimately contribute to the risk of developing neurodevelopmental disorders including ASD (Harry et al., 2006). Moreover, it has been postulated that fetal exposure to severe maternal infection, including sepsis, meningitis, and clinical chorioamnionitis, may be linked with ASD (Al-Haddad et al., 2019).

Studies examining this potential link between placental pathology and risk of ASD are few in number, with non-confirmatory results. A 2009 systematic review of eight studies found no association between placental previa, placenta abruption, infarcts, and ASD (Gardener et al., 2009). A study of 55 cases and 159 controls concluded that histological findings of acute placental inflammation were associated with higher risk of ASD among mostly early-term and preterm infants (Straughen et al., 2017). Two more recent studies, one consisting of 77 ASD cases and the other consisting of 16 ASD cases, found an association between placental pathology and ASD among preterm and extremely preterm babies, respectively (Mir et al., 2021; Raghavan et al., 2019). Generalizability of previous analyses to the broad obstetrical population is limited, as past studies had small sample sizes and included mostly infants with lower gestational ages and birthweights. Our study aims to assess the association between placental pathology and gross morphology and the risk of ASD development among both term and preterm infants.

METHODS

Study design and population

This study was based on data from a matched case–control study consisting of singleton births that occurred between 2000 and 2017 at three McGill University-affiliated hospitals. Cases were children diagnosed with confirmed ASD through the Autism Spectrum Disorders Program at the Montreal Children's Hospital. Diagnoses of ASD were confirmed through multidisciplinary assessments by hospital psychiatrists and psychologists. Controls were identified using hospital delivery logs and were matched to cases in a 1:5 ratio by hospital and date of birth; hence, allowing for control of hospital-level and time-related variation in obstetrical practice, as well as changes in diagnosis and definitions of ASD throughout time. Potential controls with known congenital malformations were excluded from the study, as were stillbirths and non-viable births of <24 weeks gestational age. This current paper focuses solely on cases and controls born at one McGill-affiliated hospital in which data on placental pathology were available.

Data collection

Maternal and newborn information was obtained from charts at hospital of birth and were entered directly into an electronic database. Abstractors were blind to the ASD diagnosis of study subjects. Maternal data variables obtained were as follows: age at delivery, pre-pregnancy weight, height, parity, gravidity, smoking history, pre-existing co-morbidities (diabetes, chronic hypertension, hypothyroidism), and clinical history (previous preterm birth, previous child with autism, and personal or family history of psychological illness). Pregnancy-related variables collected included: gestational diabetes, preeclampsia, first trimester bleeding, breech presentation, induction or augmentation of labor, mode of delivery, and fetal distress. Neonatal variables of interest included infant sex, gestational age at delivery, and birthweight.

As part of standard practice at this hospital, all placentae were reviewed by a pathologist and findings entered into a hospital-based electronic database. Hence, pathologic examinations of the placentae occurred prior to ASD diagnosis. For this study, individual pathology results were obtained from pathology records by an abstractor blind to the case–control status of the subjects. Variables of interest included: evidence of inflammation of the placenta (decidual inflammation, chorionitis, funisitis, chorioamnionitis, villitis), evidence of vasculitis (vasculitis of chorionic plate vessels, vasculitis of cord blood vessels), degree of placental maturity (immature, mature, hypermature), and other abnormalities (meconium staining, ischemic infarct, single umbilical artery, choriangioma, Tenney-Parker changes, subchorionic fibrin deposition). Tenny-Parker changes are an excessive formation of syncytial knots, which are aggregations of nuclei within the syncytioplasm of the placental epithelium (Fogarty et al., 2013). Data regarding gross morphology, namely placental weight and thickness, were also compiled, as was information regarding the length and type of insertion of the umbilical cord (eccentric, central, marginal, velamentous).

Statistical analyses

The statistical analyses were descriptive in nature. First, the maternal characteristics of cases and controls, including the characteristics of their neonates, were described. Categorical variables were presented as frequencies and percentages, and continuous variables as medians and interquartile ranges (IQR). Next, descriptive statistics were used to compare the maternal and neonatal characteristics and the placental pathologies among cases and controls; specifically, the Pearson chi-square test for categorical variables and the Wilcoxon rank-sum test for continuous variables. All analyses were conducted using R version 3.6.0. P values < 0.05 were considered to be statistically significant.

RESULTS

There were 107 cases and 526 matched controls identified at the hospital in which placental pathology reports were available. Baseline maternal and neonatal demographic and clinical characteristics of the case and control group are listed in Table 1. The mothers of the cases and controls had similar median ages at delivery, although a greater percentage of cases were 35 years or older. Mothers of cases tended to have had fewer previous births, were more commonly obese, and identified as smokers. In addition, a higher proportion of mothers of cases experienced hypothyroidism and pregnancy related morbidities, including clinical chorioamnionitis, gestational diabetes and eclampsia/preeclampsia; however, these differences between cases and controls were only statistically significant for hypothyroidism and eclampsia/preeclampsia. Compared to controls, cases were more likely to be male and born younger than 32 weeks of gestation, which were significantly different between these two groups ($p < 0.05$). Nevertheless, the median gestational ages were similar between cases and controls. Birthweights were also similar between the groups, although a greater proportion of cases were born <1500 g.

Placenta pathologies among cases and controls are compared in Table 2. An estimated 18% of births in both the case and control group had at least one placental pathology. There were essentially no differences in the rates of the individual placental pathologies between cases and controls, including placental inflammation, vasculitis, meconium staining, ischemic infarct, single umbilical artery, choriangioma, subchorionic fibrin deposition, and Tenny-Parker changes. Rates of placental maturity were similar between the two groups, although the degree of placental maturity was characterized in only 26% and 19% of births to cases and controls, respectively. An exception was placental hypermaturity, which was more common in births of infants with ASD ($p < 0.0001$).

Morphologic characteristics of the placenta are listed in Table 3. Placental thickness was similar among cases and controls, as was placental weight, and umbilical cord length and insertion. The exception was ischemic infarct, which only occurred in 2 cases and 0 controls.

DISCUSSION

The purpose of this study was to examine the relationship between placental pathology and morphology and the development of ASD. Our objective was examined using data from a matched case–control study consisting of neonates born between 2000 and 2017 at three McGill-affiliated hospitals. Cases, who were clinically-diagnosed with ASD, were matched by hospital and date of birth to controls. This current study includes data obtained at one of these sites that had pathology reports available. No differences in placental pathology and gross morphology were found between the two groups, with the exception of placental hypermaturity, which occurred more frequently among cases.

Mothers of cases and controls were overall quite similar in terms of demographic and clinical characteristics, with a few exceptions. Although the case and control groups had essentially the same median maternal age at delivery, a larger proportion of the mothers of controls were younger than 25 years of age at delivery and a greater proportion of the case group were 35 years or older at delivery. This is consistent with sibling and cohort studies that found younger maternal age to be associated with a reduced risk of ASD among offspring (Gardener et al., 2009; Parner et al., 2012). Various mechanistic hypotheses have been put forth to explain the maternal age-ASD association. For instance, a higher occurrence of spontaneous genomic mutations, an increased buildup of environmental exposures, and a higher rate of chronic conditions may explain the greater incidence of ASD among infants born to older mothers (Parner et al., 2012). Moreover, maternal obesity was more frequent among cases. Maternal obesity has been linked with the risk of ASD and is thought to impair the expression of genes involved in fetal neurogenesis (Edlow et al., 2014; Modabbernia et al., 2017). Preterm birth has also been associated with the risk of developing ASD (Agrawal et al., 2018; Leavey et al., 2013), which is consistent with our findings. Specifically, we observed that a statistically significant greater proportion of preterm babies were among cases than controls ($p < 0.05$). Shorter gestational age may disrupt the maturation of the central nervous system, thus rendering infants more vulnerable to ASD development (Leavey et al., 2013). In our study, the vast majority of children diagnosed with autism were males, which indicates a sex-related differential in ASD (Werling, 2016). We also found a significant difference in frequency of hypothyroidism between cases and controls, with maternal hypothyroidism being more

TABLE 1 Maternal and neonatal characteristics of the study population

	Cases (total = 107) *n* (%)[a]	Controls (total = 526) *n* (%)[a]
Maternal characteristics		
Maternal age at delivery (years), median (IQR)	31.0 [29, 35]	31.5 [28, 35]
Maternal age at delivery (years)		
<25	5 (4.7)	58 (11.0)
25–34	67 (62.6)	330 (62.7)
≥35	35 (32.7)	138 (26.2)
Parity		
0	52 (48.6)	213 (40.5)
1	34 (31. 8)	182 (34.6)
2	12 (11.2)	70 (13.3)
≥3	9 (8.4)	61 (11.6)
Gravidity		
1	34 (31.8)	157 (29.9)
≥2	73 (68.2)	369 (70.2)
Pre-pregnancy maternal BMI, kg/m^2		
Underweight or normal (<25.0)	39 (36.4)	195 (37.1)
Overweight (25.0–29.9)	14 (13.1)	91 (17.3)
Obese (30.0 or more)	17 (15.9)	53 (10.1)
Missing	37 (34.6)	187 (35.6)
Smoking	6 (5.6)	14 (2.7)
Comorbidities		
Chronic hypertension	1 (0.9)	6 (1.1)
Hypothyroidism[b]	13 (12.2)	28 (5.3)
Diabetes	1 (0.9)	5 (0.95)
Pregnancy-related morbidities		
Clinical chorioamnionitis	12 (11.2)	42 (8.0)
Gestational diabetes	10 (9.4)	45 (8.6)
Preeclampsia or eclampsia[b]	8 (7.5)	11 (2.1)
Neonatal characteristics		
Gestational age, weeks[b]		
<32	6 (5.6)	5 (1.0)
32–36	7 (6.5)	32 (6.1)
37–39	58 (54.2)	295 (56.2)
≥40	36 (33.6)	193 (36.7)
Median gestational age, weeks (IQR)	39.3 [38.6, 40.1]	39.6 [38.6, 40.3]
Male sex[b]	83 (77.6)	273 (51.9)
Median birth weight, g (IQR)	3380 [2995, 3755]	3370 [3050, 3675]
Birth weight, g[b]		
<1500	6 (5.6)	4 (0.8)
1500–2500	6 (5.6)	23 (4.4)
2500–3500	50 (46.7)	303 (57.6)
>3500	45 (42.1)	196 (37.3)

[a]Cell totals and percentages are presented unless otherwise noted.
[b]*p*-value <0.05. The other variables had *p*-values >0.05.

common among women with offspring diagnosed with ASD. This is in concordance with a recent meta-analysis that found that maternal thyroid dysfunction was related to offspring neurodevelopment. Specifically, they observed that maternal hypothyroidism was associated with a 41% greater likelihood of having an infant diagnosed with ASD (OR 1.41, 95% CI 1.05–1.90 (Ge et al., 2020). Preeclampsia was also more common among cases in our study; a finding that has been observed in the current literature (Barron et al., 2021). Although the mechanism responsible for this relationship is unknown, Barron et al postulate that inflammation and oxidative stress, which are characteristic of preeclampsia, may influence this association.

Past studies have linked intrauterine inflammation and abnormal placentation with the risk of cerebral palsy, schizophrenia, psychomotor impairment (Goldstein et al., 2020; Suvisaari et al., 2013; Xiao et al., 2018; Xing et al., 2019). It has been proposed that cytokines and other inflammatory agents could disrupt white matter maturation and contribute to the development of neurodevelopmental disorders such as ASD (Harry et al., 2006). With regard to chorioamnionitis, only a handful of studies have evaluated the potential effect of this condition on the risk of ASD, with findings differing by whether term and/or preterm infants were examined. While previous studies of preterm infants reported positive associations between histologic chorioamnionitis and ASD (Limperopoulos et al., 2008; Venkatesh et al., 2020), analyses including both term and preterm births showed conflicting results (Raghavan et al., 2019; Straughen et al., 2017). One study that included both term and preterm infants, found an association between histological findings of placental inflammation and the risk of ASD (Straughen et al., 2017). Another study reported that chorioamnionitis was associated with the risk of ASD in spontaneous preterm infants, but not in term infants (Raghavan et al., 2019). The lack of a statistically significant difference in frequency of histologic chorioamnionitis between cases and controls in our study may be attributable to the characteristics of our sample. Specifically, in our study, only 12% of cases and 7% controls were born <37 weeks gestational age. The one study that reported an association between placental inflammation and ASD in a cohort of term and preterm infants had a lower mean gestational

TABLE 2 Placental pathology among cases and controls

	Cases (total = 107) *n* (%)[a]	Controls (total = 526) *n* (%)[a]	*p*-value[b]
Any placental pathology[c]	19 (17.8)	94 (17.9)	0.97
Placental inflammation	8 (7.5)	42 (8.0)	0.90
Decidual inflammation	0 (0)	1 (0.2)	0.65
Chorionitis	1 (0.9)	13 (2.5)	0.33
Funisitis	5 (4.7)	19 (3.6)	0.59
Chorioamnionitis	4 (3.7)	26 (4.9)	0.60
Villitis	1 (1.1)	2 (0.4)	0.44
Vasculitis			
Vasculitis of chorionic plate vessels	3 (2.8)	15 (2.9)	0.99
Vasculitis of cord blood vessels	4 (4.7)	11 (2.6)	0.31
Degree of placental maturity			
Immature	1 (1.8)	4 (1.5)	0.85
Mature	22 (20.6)	96 (18.3)	0.59
Hypermature	5 (4.7)	2 (0.4)	<0.0001
Other pathologies			
Meconium staining	8 (7.7)	61 (12)	0.21
Ischemic infarct	2 (1.9)	0 (0)	0.03
Single umbilical artery	1 (1.3)	4 (0.8)	0.68
Choriangioma	1 (1.3)	2 (0.5)	0.44
Tenney-Parker changes	1 (1.0)	1 (0.2)	0.21
Subchorionic fibrin deposition	1 (1.0)	1 (0.2)	0.21

[a]Cell totals and percentages are presented unless otherwise noted.
[b]From Pearson χ^2-test.
[c]Defined as having one or more pathologies listed in this table.

TABLE 3 Placental gross morphology among cases and controls

	Cases (total = 107) *n* (%)[a]	Controls (total = 526) *n* (%)[a]	*p*-value[b]
Placental weight (g), median (IQR)	590 [500, 650]	580 [510, 680]	0.51
Average placental thickness (cm), median (IQR)	1.7 [1.4, 2.2]	1.7 [1.4, 2.3]	0.56
Cord length (cm), median (IQR)	50 [38, 60]	51 [42, 60]	0.16
Cord insertion			
Eccentric	69 (64.5)	361 (68.6)	0.68
Central	23 (21.5)	87 (16.5)	
Marginal	12 (11.2)	63 (12.0)	
Velamentous	3 (2.8)	15 (2.9)	

[a]Cell totals and percentages are presented unless otherwise noted.
[b]From Pearson χ^2-test or Wilcoxon rank-sum test, as appropriate.

age in their sample (37.6 weeks) compared to our study (39.5 weeks) (Straughen et al., 2017). In preterm and early-term infants, chorioamnionitis is associated with more severe inflammation of the fetal membranes, including the amnion and umbilical cord (Conti et al., 2015; Kim et al., 2001). Conversely, histologic chorioamnionitis in term births is believed to primarily affect maternal surfaces, such as the decidua and chorionic plate, and result in limited fetal inflammatory response (Conti et al., 2015; Raghavan et al., 2019; Rodríguez-Trujillo et al., 2019). Although the results of our study suggest that histologic chorioamnionitis is not associated with development of ASD, based on the previous literature, which is inconclusive, we recommend further research to examine the potential association between chorioamnionitis and ASD.

In this current study, placental hypermaturity was more common among cases. A previous study found placental immaturity was associated with ASD cases, although the relationship was not statistically significant (Raghavan et al., 2019). While there is a paucity of evidence on the relation of placental maturation and the risk of ASD, placental hypermaturity has been linked with poor vascular perfusion and may contribute to fetal hypoxic insults (Turowski & Vogel, 2018).

To our knowledge, our study is the second contemporary analysis to evaluate the effect of placental vasculitis on the risk of ASD (Straughen et al., 2017). Abnormal vascularization has been linked with low placental efficiency (Salafia et al., 2006; Salafia et al., 2012), which causes a reduced fetal supply of gases and nutrients and ultimately results in impaired development of the central nervous system and increased risk of ASD development (Prado & Dewey, 2014). A case–control study by Straughen et al suggests that vasculitis of chorionic plate vessel is associated with a 5.1-fold increase in the risk of ASD among preterm and term infants (Straughen et al., 2017). In contrast, we found no differences in risk of vasculitis between cases and controls. Interestingly, previous studies have found that high rates of antenatal steroid use may reduce intrauterine inflammation (Been et al., 2011). Given the predominance of our newborns being term, steroids were not commonly given.

There are indications in the literature that reduced placental thickness, as well as marginal and velamentous cord insertions, could reflect impaired development and function of the placenta (Buchanan-Hughes et al., 2020; Ismail et al., 2017; Salafia et al., 2006; Vahanian et al., 2015). Reduced placental thickness has been linked with inadequate arborization of the villous tree and decreased vascular nutrient exchange surface (Salafia et al., 2006). In addition, non-central cord insertions have been associated with umbilical vessel compression and rupture (Buchanan-Hughes et al., 2020; Ismail et al., 2017), low placental efficiency, and increased risks of perinatal deaths and preterm births (Vahanian et al., 2015). However, there is a paucity of evidence to support the associations of placental thickness and cord insertion with risk of ASD development. A study comparing placental disk thickness of 129 high-risk ASD births and 267 controls found no differences between the two groups (Park et al., 2018). Another study of 51 high-risk ASD births and 51 controls found that cases had reduced placental thickness, and increased rates of marginal cord insertion, when compared to controls (Salafia et al., 2012). We found that average placental thickness and sites of cord insertion were not different between cases and controls. Other investigations have assessed placental thickness and umbilical insertions among high-risk ASD cases, rather than infants with confirmed ASD diagnosis, which may explain their observed effect (Chen et al., 2020; Park et al., 2018; Salafia et al., 2012). Moreover, risk of ASD may be related to different perinatal factors, such as gestational age, parental age and maternal morbidity, which might also influence placental thickness (Baptiste-Roberts et al., 2008).

The main limitations of our study are due to its relatively small sample size; nonetheless, this is one of the largest studies of validated cases of ASD evaluating placental pathologies. One limitation was the inability to estimate measures of effect between placental pathologies and gross morphologic differences and ASD risk; instead, we were limited to performing univariate analyses. In addition, given the low rate of pathologies, regression analyses were not felt to be of added value and thus, were not performed. This also rendered us unable to consider in our analyses any imbalances in frequency of covariates between the case and control groups. Further, overall, the low incidence of placental pathologies resulted in conclusions being made on few subjects. Hence, overall, study results should be interpreted cautiously. Further, we also lacked data on variables that would have allowed us to more fully examine our study objective and to better interpret the study findings, such as specific type of autism disorder or disease severity. Lastly, we lacked data on placental trophoblast inclusions that may reflect a genetic vulnerability to neurodevelopmental disorders. Specifically, a significant increase in trophoblast inclusions have been associated with greater risk of autism (Anderson et al., 2007; Walker et al., 2013).

Despite these drawbacks, our study had some strengths to highlight. First, the cases in our study received validated, confirmed diagnoses of ASD, rather than just being identified through an existing database. Second, we were able to obtain data pertaining to maternal demographic and comorbid conditions, and also detailed information about pathological and morphological conditions of the placentae from the medical charts and reports. Further, placentae were assessed for pathologic and morphologic abnormalities at birth, prior to diagnoses of ASD, hence, preventing potential differential misclassification. To our knowledge, this is the largest contemporary study to examine the effect of placental histopathology on the risk of ASD among both term and preterm births.

Our study showed no association between placental pathology and gross morphology with the risk of ASD development, which suggests that placental abnormalities are unlikely to either cause or mitigate the development of ASD. However, due to our study limitations, further research should be conducted to clarify the relationship between placental pathology and gross morphology and ASD before any definitive conclusions can be drawn.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interest.

ETHICS STATEMENT

Ethics approval was obtained for this study by the MBM research Ethics Committee of the Jewish General Hospital (MM-CODIM-MBM-CR17-53).

DATA AVAILABILITY STATEMENT

Author elects to not share data.

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REFERENCES

Agrawal, S., Rao, S. C., Bulsara, M. K., & Patole, S. K. (2018). Prevalence of autism spectrum disorder in preterm infants: A meta-analysis. *Pediatrics*, *142*, e20180134. https://doi.org/10.1542/peds.2018-0134

Al-Haddad, B. J., Jacobsson, B., Chabra, S., Modzelewska, D., Olson, E. M., Bernier, R., Enquobahrie, D. A., Hagberg, H., Östling, S., & Rajagopal, L. (2019). Long-term risk of neuropsychiatric disease after exposure to infection in utero. *JAMA Psychiatry*, *76*, 594–602.

Anderson, G. M., Jacobs-Stannard, A., Chawarska, K., Volkmar, F. R., & Kliman, H. J. (2007). Placental trophoblast inclusions in autism spectrum disorder. *Biological Psychiatry*, *61*, 487–491. https://doi.org/10.1016/j.biopsych.2006.03.068

Armstrong-Wells, J., Donnelly, M., Post, M. D., Manco-Johnson, M. J., Winn, V. D., & Sébire, G. (2015). Inflammatory predictors of neurologic disability after preterm premature rupture of membranes. *American Journal of Obstetrics and Gynecology*, *212*, 212.e211–219. https://doi.org/10.1016/j.ajog.2014.09.016

Baptiste-Roberts, K., Salafia, C. M., Nicholson, W. K., Duggan, A., Wang, N.-Y., & Brancati, F. L. (2008). Maternal risk factors for abnormal placental growth: The national collaborative perinatal project. *BMC Pregnancy and Childbirth*, *8*, 44.

Barron, A., McCarthy, C. M., & O'Keeffe, G. W. (2021). Preeclampsia and neurodevelopmental outcomes: Potential pathogenic roles for inflammation and oxidative stress? *Molecular Neurobiology*, *58*, 2734–2756. https://doi.org/10.1007/s12035-021-02290-4

Been, J. V., Degraeuwe, P. L., Kramer, B. W., & Zimmermann, L. J. (2011). Antenatal steroids and neonatal outcome after chorioamnionitis: A meta-analysis. *BJOG*, *118*, 113–122. https://doi.org/10.1111/j.1471-0528.2010.02751.x

Buchanan-Hughes, A., Bobrowska, A., Visintin, C., Attilakos, G., & Marshall, J. (2020). Velamentous cord insertion: Results from a rapid review of incidence, risk factors, adverse outcomes and screening. *Systematic Reviews*, *9*, 147. https://doi.org/10.1186/s13643-020-01355-0

Chen, S., Chen, C., Jan, M., Feng, J. S., Krickellas, J., Chowdhury, S. F., Nantwi, A., Dygulski, S., Bromberg, H., Shah, R., Pongsachai, J., Joyce, M., Dygulska, B., & Salafia, C. (2020). Placental gross pathology in a population based case-control study of autism spectrum disorder. *Reproductive Sciences*, *27*, 147A.

Conti, N., Torricelli, M., Voltolini, C., Vannuccini, S., Clifton, V. L., Bloise, E., & Petraglia, F. (2015). Term histologic chorioamnionitis: A heterogeneous condition. *European Journal of Obstetrics & Gynecology and Reproductive Biology*, *188*, 34–38.

Edlow, A. G., Vora, N. L., Hui, L., Wick, H. C., Cowan, J. M., & Bianchi, D. W. (2014). Maternal obesity affects fetal neurodevelopmental and metabolic gene expression: A pilot study. *PLoS One*, *9*, e88661. https://doi.org/10.1371/journal.pone.0088661

Elsabbagh, M., Divan, G., Koh, Y. J., Kim, Y. S., Kauchali, S., Marcín, C., Montiel-Nava, C., Patel, V., Paula, C. S., & Wang, C. (2012). Global prevalence of autism and other pervasive developmental disorders. *Autism Research*, *5*, 160–179.

Fogarty, N. M., Ferguson-Smith, A. C., & Burton, G. J. (2013). Syncytial knots (Tenney-Parker changes) in the human placenta: Evidence of loss of transcriptional activity and oxidative damage. *The American Journal of Pathology*, *183*, 144–152. https://doi.org/10.1016/j.ajpath.2013.03.016

Gardener, H., Spiegelman, D., & Buka, S. L. (2009). Prenatal risk factors for autism: Comprehensive meta-analysis. *The British Journal of Psychiatry*, *195*, 7–14.

Gardener, H., Spiegelman, D., & Buka, S. L. (2011). Perinatal and neonatal risk factors for autism: A comprehensive meta-analysis. *Pediatrics*, *128*, 344–355.

Ge, G. M., Leung, M. T. Y., Man, K. K. C., Leung, W. C., Ip, P., Li, G. H. Y., Wong, I. C. K., Kung, A. W. C., & Cheung, C. L. (2020). Maternal thyroid dysfunction during pregnancy and the risk of adverse outcomes in the offspring: A systematic review and meta-analysis. *The Journal of Clinical Endocrinology and Metabolism*, *105*, 3821–3841. https://doi.org/10.1210/clinem/dgaa555

Goldstein, J. A., Gallagher, K., Beck, C., Kumar, R., & Gernand, A. D. (2020). Maternal-fetal inflammation in the placenta and the developmental origins of health and disease. *Frontiers in Immunology*, *11*, 531543. https://doi.org/10.3389/fimmu.2020.531543

Harry, G. J., Lawler, C., & Brunssen, S. H. (2006). Maternal infection and white matter toxicity. *Neurotoxicology*, *27*, 658–670. https://doi.org/10.1016/j.neuro.2006.05.004

Ismail, K. I., Hannigan, A., O'Donoghue, K., & Cotter, A. (2017). Abnormal placental cord insertion and adverse pregnancy outcomes: A systematic review and meta-analysis. *Systematic Reviews*, *6*, 242. https://doi.org/10.1186/s13643-017-0641-1

Kim, C. J., Yoon, B. H., Park, S.-S., Kim, M. H., & Chi, J. G. (2001). Acute funisitis of preterm but not term placentas is associated with severe fetal inflammatory response. *Human Pathology*, *32*, 623–629.

Kuypers, E., Ophelders, D., Jellema, R. K., Kunzmann, S., Gavilanes, A. W., & Kramer, B. W. (2012). White matter injury following fetal inflammatory response syndrome induced by chorioamnionitis and fetal sepsis: Lessons from experimental ovine models. *Early Human Development*, *88*, 931–936.

Leavey, A., Zwaigenbaum, L., Heavner, K., & Burstyn, I. (2013). Gestational age at birth and risk of autism spectrum disorders in Alberta, Canada. *The Journal of Pediatrics*, *162*, 361–368. https://doi.org/10.1016/j.jpeds.2012.07.040

Limperopoulos, C., Bassan, H., Sullivan, N. R., Soul, J. S., Robertson, R. L., Jr., Moore, M., Ringer, S. A., Volpe, J. J., & Plessis, A. J. D. (2008). Positive screening for autism in ex-preterm infants: Prevalence and risk factors. *Pediatrics*, *121*, 758–765. http://ovidsp.ovid.com/ovidweb.cgi?T=JS&CSC=Y&NEWS=N&PAGE=fulltext&D=emed10&AN=354416823, http://mcgill.on.worldcat.org/atoztitles/link?sid=OVID:embase&id=pmid:18381541&id=doi:10.1542%2Fpeds.2007-2158&issn=0031-4005&isbn=&volume=121&issue=4&spage=758&pages=758-765&date=2008&title=Pediatrics&atitle=Positive+screening+for+autism+in+ex-preterm+infants%3A+Prevalence+and+risk+factors&aulast=Limperopoulos

Mir, I. N., White, S. P., Steven Brown, L., Heyne, R., Rosenfeld, C. R., & Chalak, L. F. (2021). Autism spectrum disorders in extremely preterm infants and placental pathology findings: A matched case-control study. *Pediatric Research*, *89*, 1825–1831. https://doi.org/10.1038/s41390-020-01160-4

Modabbernia, A., Velthorst, E., & Reichenberg, A. (2017). Environmental risk factors for autism: An evidence-based review of systematic reviews and meta-analyses. *Molecular Autism*, *8*, 13. https://doi.org/10.1186/s13229-017-0121-4

National Center on Birth Defects and Developmental Disabilities, Centers for Disease Control and Prevention. (2021). *Data and statistics* *on autism spectrum disorder*. https://www.cdc.gov/ncbddd/autism/data.html.

Park, B. Y., Misra, D. P., Moye, J., Miller, R. K., Croen, L., Fallin, M. D., Walker, C., Newschaffer, C. J., Salafia, C. M., & Natl Childrens Study, C. (2018). Placental gross shape differences in a high autism risk cohort and the general population. *PLoS One*, *13*, e0191276. https://doi.org/10.1371/journal.pone.0191276

Parner, E. T., Baron-Cohen, S., Lauritsen, M. B., Jørgensen, M., Schieve, L. A., Yeargin-Allsopp, M., & Obel, C. (2012). Parental age and autism spectrum disorders. *Annals of Epidemiology*, *22*, 143–150. https://doi.org/10.1016/j.annepidem.2011.12.006

Prado, E. L., & Dewey, K. G. (2014). Nutrition and brain development in early life. *Nutrition Reviews*, *72*, 267–284. https://doi.org/10.1111/nure.12102

Raghavan, R., Helfrich, B. B., Cerda, S. R., Ji, Y., Burd, I., Wang, G., Hong, X., Fu, L., Pearson, C., Daniele Fallin, M., Zuckerman, B., & Wang, X. (2019). Preterm birth subtypes, placental pathology findings, and risk of neurodevelopmental disabilities during childhood [Research Support, N.I.H., Extramural Research Support, Non-U.S. Gov't Research Support, U.S. Gov't, P.H.S.]. *Placenta*, *83*, 17–25. https://doi.org/10.1016/j.placenta.2019.06.374

Rodríguez-Trujillo, A., Ríos, J., Ángeles, M. A., Posadas, D. E., Murillo, C., Rueda, C., Botet, F., Bosch, J., Vergara, A., & Gratacós, E. (2019). Influence of perinatal inflammation on the neurodevelopmental outcome of premature infants. *The Journal of Maternal-Fetal & Neonatal Medicine*, *32*, 1069–1077.

Ronald, A., & Hoekstra, R. A. (2011). Autism spectrum disorders and autistic traits: A decade of new twin studies. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics*, *156*, 255–274.

Salafia, C., Misra, D., Golding, J., Platt, C., & Ring, S. (2012). Characterization of placental growth as a biomarker of autism/ASD risk. *Placenta*, *33*, A16.

Salafia, C. M., Charles, A. K., & Maas, E. M. (2006). Placenta and fetal growth restriction. *Clinical Obstetrics and Gynecology*, *49*, 236–256.

Straughen, J. K., Misra, D. P., Divine, G., Shah, R., Perez, G., VanHorn, S., Onbreyt, V., Dygulska, B., Schmitt, R., Lederman, S., Narula, P., & Salafia, C. M. (2017). The association between placental histopathology and autism spectrum disorder. *Placenta*, *57*, 183–188. https://doi.org/10.1016/j.placenta.2017.07.006

Suvisaari, J., Taxell-Lassas, V., Pankakoski, M., Haukka, J., Lonnqvist, J., & Hakkinen, L. (2013). Obstetric complications as risk factors for schizophrenia spectrum psychoses in offspring of mothers with psychotic disorder [Conference Abstract]. *Schizophrenia Bulletin*, *39*, S79. http://ovidsp.ovid.com/ovidweb.cgi?T=JS&CSC=Y&NEWS=N&PAGE=fulltext&D=emed14&AN=71120255, http://mcgill.on.worldcat.org/atoztitles/link?sid=OVID:embase&id=pmid:&id=doi:10.1093%2Fschbul%2Fsbt011&issn=0586-7614&isbn=&volume=39&issue=SUPPL.+1&spage=S79&pages=S79&date=2013&title=Schizophrenia+Bulletin&atitle=Obstetric+complications+as+risk+factors+for+schizophrenia+spectrum+psychoses+in+offspring+of+mothers+with+psychotic+disorder&aulast=Suvisaari

Tomlinson, M. S., Lu, K., Stewart, J. R., Marsit, C. J., O'Shea, T. M., & Fry, R. C. (2019). Microorganisms in the placenta: Links to early-life inflammation and neurodevelopment in children. *Clinical Microbiology Reviews*, *32*, e00103-18. https://doi.org/10.1128/cmr.00103-18

Turowski, G., & Vogel, M. (2018). Re-view and view on maturation disorders in the placenta. *APMIS*, *126*, 602–612.

Vahanian, S. A., Lavery, J. A., Ananth, C. V., & Vintzileos, A. (2015). Placental implantation abnormalities and risk of preterm delivery: A systematic review and metaanalysis. *American Journal of Obstetrics and Gynecology*, *213*, S78–S90. https://doi.org/10.1016/j.ajog.2015.05.058

van Vliet, E. O., de Kieviet, J. F., van der Voorn, J. P., Been, J. V., Oosterlaan, J., & van Elburg, R. M. (2012). Placental pathology and long-term neurodevelopment of very preterm infants. *American Journal of Obstetrics and Gynecology*, *206*, 489. e481–489. e487.

Venkatesh, K. K., Leviton, A., Hecht, J. L., Joseph, R. M., Douglass, L. M., Frazier, J. A., Daniels, J. L., Fry, R. C., O'Shea, T. M., & Kuban, K. C. (2020). Histologic chorioamnionitis and risk of neurodevelopmental impairment at age 10 years among extremely preterm infants born less than 28 weeks of gestation. *American Journal of Obstetrics & Gynecology*, *06*, 745.e10. https://doi.org/10.1016/j.ajog.2020.05.001

Walker, C. K., Anderson, K. W., Milano, K. M., Ye, S., Tancredi, D. J., Pessah, I. N., Hertz-Picciotto, I., & Kliman, H. J. (2013). Trophoblast inclusions are significantly increased in the placentas of children in families at risk for autism. *Biological Psychiatry*, *74*, 204–211. https://doi.org/10.1016/j.biopsych.2013.03.006

Werling, D. M. (2016). The role of sex-differential biology in risk for autism spectrum disorder. *Biology of Sex Differences*, *7*, 58. https://doi.org/10.1186/s13293-016-0112-8

Xiao, D., Zhu, T., Qu, Y., Gou, X., Huang, Q., Li, X., & Mu, D. (2018). Maternal chorioamnionitis and neurodevelopmental outcomes in preterm and very preterm neonates: A meta-analysis. *PLoS One*, *13*, e0208302.

Xing, L., Wang, G., Chen, R., Ren, J., Qian, J., & Huang, Y. (2019). Is chorioamnionitis associated with neurodevelopmental outcomes in preterm infants? A systematic review and meta-analysis following PRISMA. *Medicine*, *98*, e18229.

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